EX.No.2.1 – CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

Date:12.10.2026

AIM:

To construct a three-dimensional solid model from the given isometric drawing using Autodesk Fusion 360 software.

REQUIREMENTS:

Autodesk Fusion 360 software
Windows 10 operating system
16 GB RAM
Integrated graphics with 1 GB or more recommended
Internet connection with 2.5 Mbps download and 500 Kbps upload speed or higher

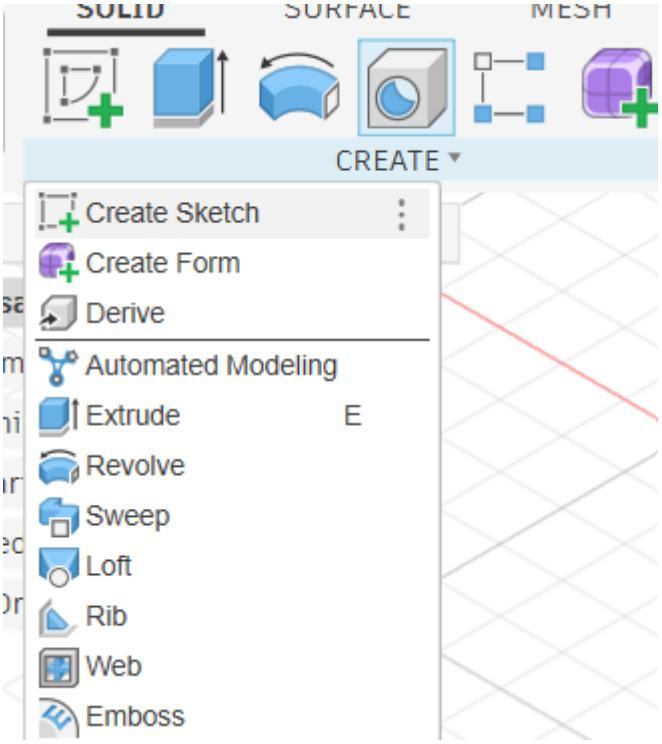
PROCEDURE:

1: Open the Modelling Workspace

Launch Autodesk Fusion 360 and create a new design. Ensure that the workspace is set to Design.

2: Prepare the Base Profile

Click Create Sketch and choose the XZ plane. Use the Rectangle tool to draw the base profile with a width of 100 mm and a height of 60 mm.



3: Apply Dimensions

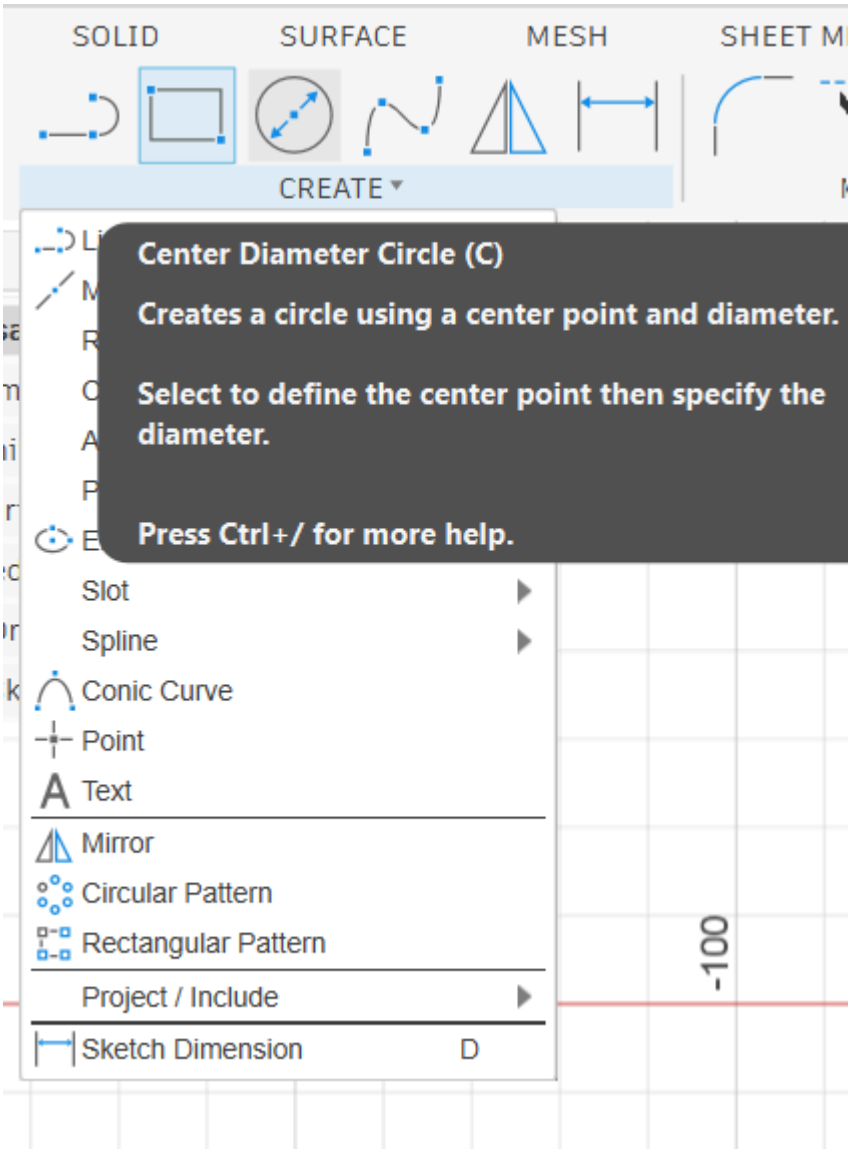
Use the Sketch Dimension command to specify the required width and height. Check the dimensions before completing the sketch.

4:Form the base of solid

Select the closed rectangular profile and click Extrude. Set the extrusion distance to 30 mm and confirm to produce the base solid.

5: Add a Circular Feature

Select the front face of the rectangular solid and create a new sketch. Choose the Center Diameter Circle tool and draw the circle according to the given isometric drawing.



6: Specify the Circle Size

Apply the required diameter using the dimension command. Position the circle correctly on the front face and click Finish Sketch.

7: Build the Circular Projection

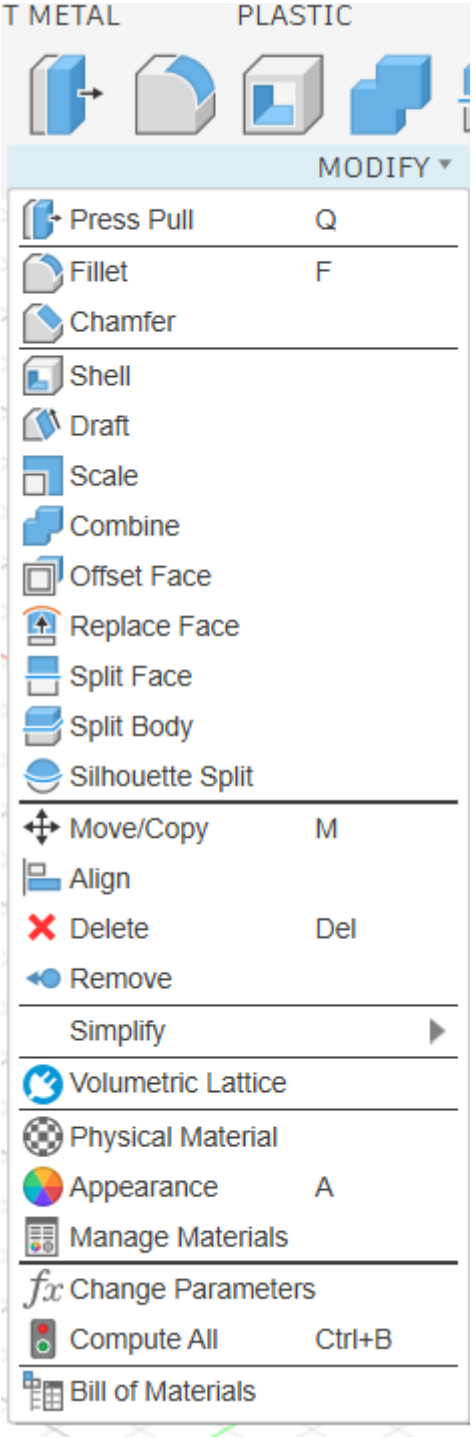
Select the circular profile and activate the Extrude command. Enter the required distance of 20 mm and choose the appropriate operation to form the circular feature.

8:Sketch on the top face

Select the top face of the base and start another sketch. Use the Line tool to draw the additional edges and profiles shown in the reference drawing.

9: Modify the Model

Finish the sketch and apply the necessary extrusion or cut operation to obtain the required shape. Check that the new features align correctly with the base solid.



10: Verify the Final Model

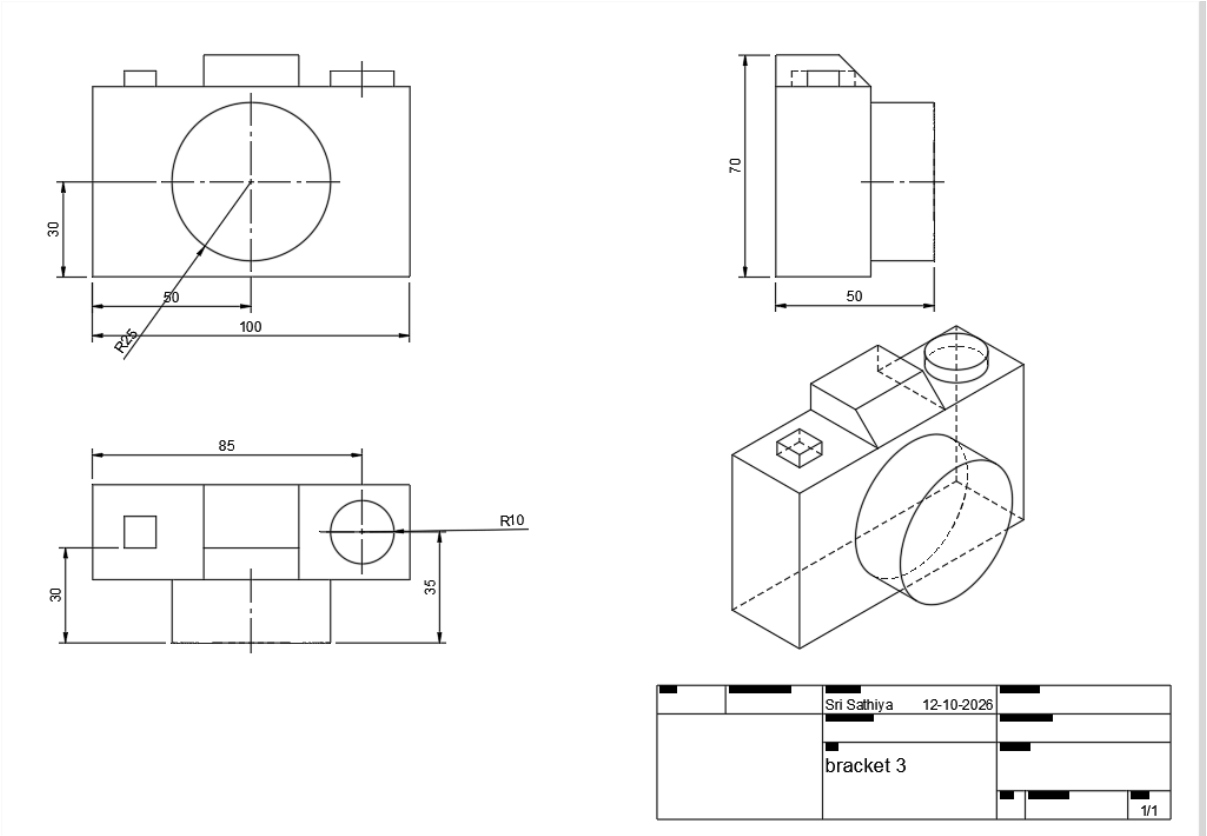
Rotate the model using the ViewCube and inspect it from the front, top, and isometric views. Confirm that the dimensions and features match the given drawing.

11: Save the Design

Click Save, enter a suitable file name, and store the completed 3D model.

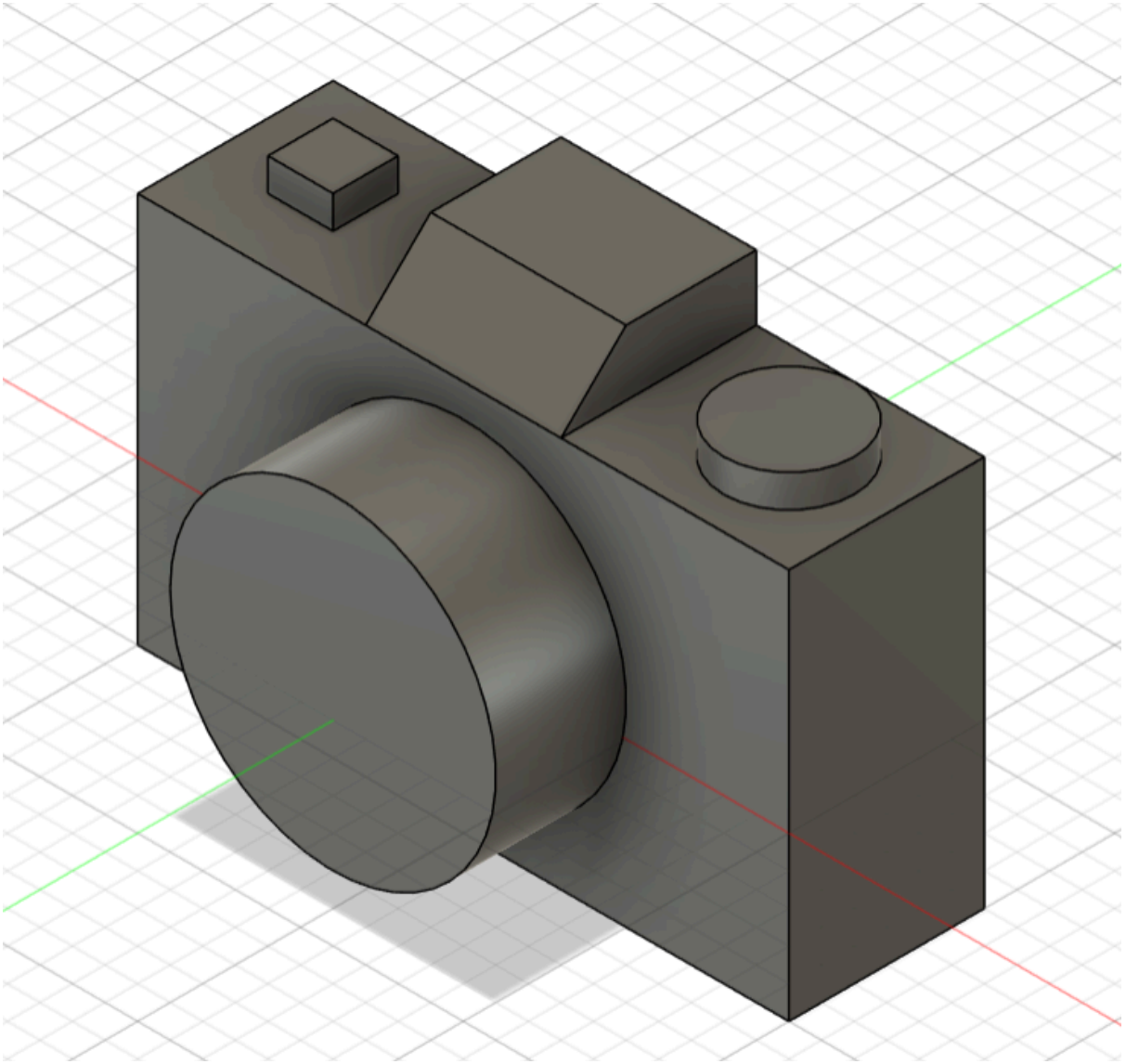
INPUT:

Attach the given isometric drawing.



OUTPUT:

Attach the final 3D solid model created using Autodesk Fusion 360.



RESULT:

Thus, the given isometric drawing was modelled as a 3D solid using Autodesk Fusion 360 software.